

SKILL – 7

Code:

```
GNU nano 7.2 skill7_waitpid.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    int status;

    printf("=====\n");
    printf(" SKILL 7 - PROCESS SYNCHRONIZATION\n");
    printf("=====\n");

    printf("Parent Process PID: %d\n", getpid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {

```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-07$ ./skill7_waitpid
=====  
SKILL 7 - PROCESS SYNCHRONIZATION  
=====  
Parent Process PID: 452  
  
Parent is waiting for Child...  
Child PID: 453  
  
Child Process Created  
Child PID: 453  
Child is executing...  
Child Process Completed  
  
Child process terminated normally.  
Child Exit Status: 0  
Parent Process Completed  
=====
```

Check waitpid in the source code:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-07$ nano skill7_waitpid.c  
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-07$ ls -l skill7_waitpid  
-rwxr-xr-x 1 bhardwaj bhardwaj 16320 Oct  7 03:38 skill7_waitpid  
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-07$ ls -l skill7_waitpid  
-rwxr-xr-x 1 bhardwaj bhardwaj 16320 Oct  7 03:38 skill7_waitpid  
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-07$ grep -n "waitpid" skill7_waitpid.c  
41:         waitpid(pid, &status, 0);  
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-07$ |
```

Check fork():

```
41:         waitpid(pid, &status, 0);  
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-07$ grep -n "fork" skill7_waitpid.c  
17:     pid = fork();  
21:         perror("fork failed");  
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-07$ |
```

Check process IDs:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-07$ ps -f  
UID          PID     PPID  C  STIME TTY          TIME CMD  
bhardwaj      287       283  0  03:35 pts/0        00:00:00 -bash  
bhardwaj      480       287  0  03:42 pts/0        00:00:00 ps -f  
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-07$ |
```